



SECTION VI

PARTS LISTING

6.1 GENERAL

This section contains a list of all replaceable parts. The section also provides a means to locate desired components. Table 6-1 lists the various components and their part numbers and descriptions. Figure 6-1 shows the layout of the Interval Timer circuit board with the location of the various integrated circuits, capacitors, resistors and the crystal.

Table 6-1. Interval Timer Parts List

Reference Number	Part Number	Description
C1, C2	230855-0001	Capacitor, 68 mF, 15 V, 10%, case size C
C3-C12 C15, C16 C17	230590-9000	Capacitor, 0.05 mF, 12 V, 20%, cer. transcap
C13	230426-0000	Capacitor, 150 pF, 500 V
C14	230381-0000	Capacitor, 22.00 pF, 500 V
R1, R2	233650-0000	Resistor, 470 ohm, 0.25 W
R3-R6	233756-0000	Resistor, 1000 ohm, 0.25 W
S1	972547-0007	Switch, slide-SPST, DIP 7 switches
Y1	226591-0003	Crystal unit, quartz, 10.000 MHz, +0.005%
Z1, Z3, Z5, Z6	222222-7173	IC network SN74173N
Z2, Z4, Z13, Z24	240000-7404	IC network SN74H04N
Z7, Z9, Z11	222222-7163	IC network SN74163N
Z8	222222-7174	IC network SN74174N
Z10, Z16 Z20	222222-7175	IC network SN74175N
Z12, Z18, Z25, Z32	240000-7103	IC network SN74H103N



Table 6-1. Interval Timer Parts List (Continued)

Reference Number	Part Number	Description
Z14	222222-7485	IC network SN7485N
Z15	219402-7133	IC network SN74S133N
Z17, Z23, Z38	222222-7400	IC network SN7400N
Z19, Z26, Z27, Z33, Z34	222222-7162	IC network SN74162N
Z21	240000-7474	IC network SN74H74N
Z22	240000-7410	IC network SN74H10N
Z28	222222-7437	IC network SN7437N
Z29	240000-7411	IC network SN74H11N
Z30	222222-7402	IC network SN7402N
Z31, Z37	222222-7410	IC network SN7410N
Z35	222222-7405	IC network SN7405N
Z36	222222-7408	IC network SN7408N
Z39	222222-7420	IC network SN7420N



943001-9701

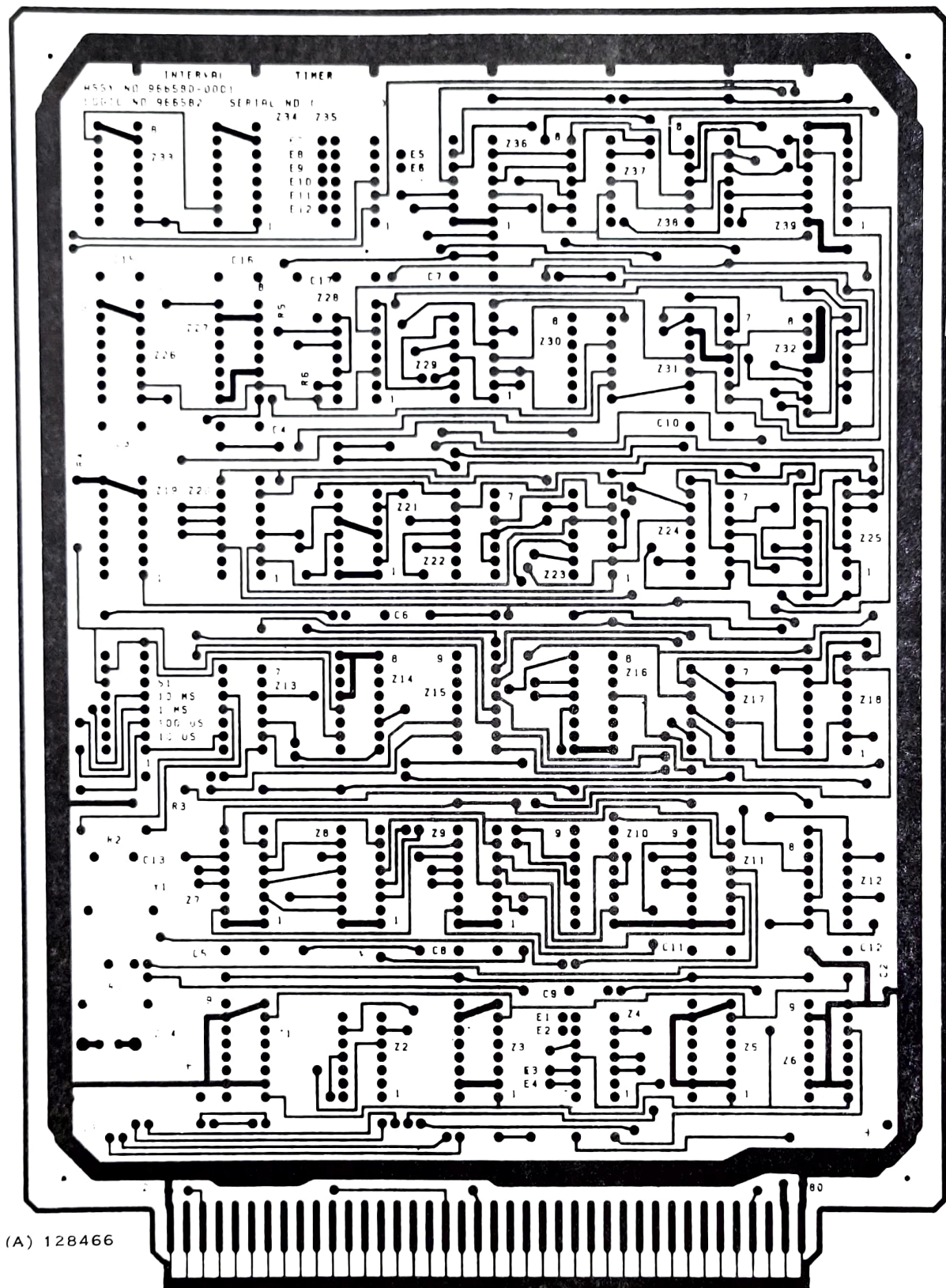


Figure 6-1. Interval Timer Parts Layout





SECTION VII

DIAGRAMS

7.1 GENERAL

This section contains the logic diagrams and applicable drawings for the 980A Interval Timer.

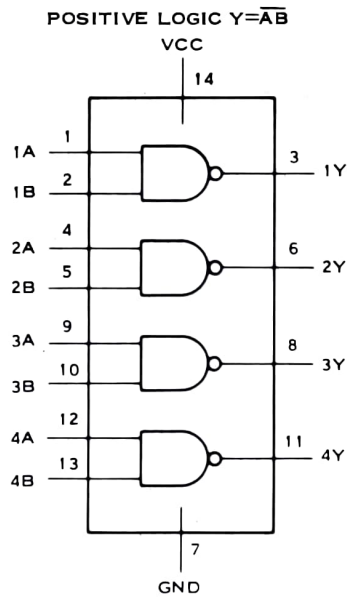
7.2 INTEGRATED CIRCUIT DATA

Figure 7-1 contains diagrams for the different integrated circuits found in the Interval Timer. These diagrams can be used with the logic diagrams to better understand the circuits.

7.3 LIST OF DRAWINGS

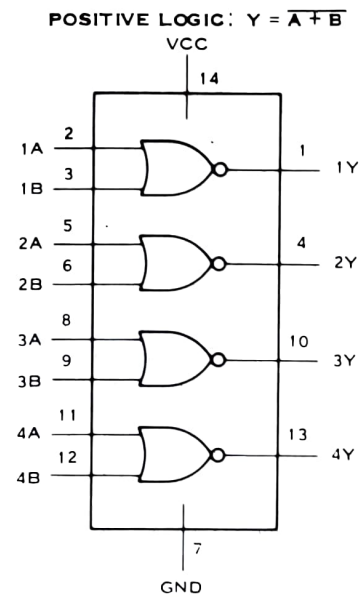
The Interval Timer drawings follow the integrated circuit diagrams and are:

<u>Name</u>	<u>Drawing Number</u>
Detailed Logic Diagram	966582 (2 sheets)
Interval Timer, 980A	966580



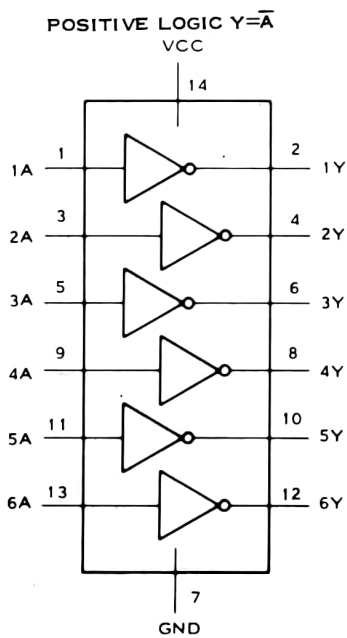
SN7400

QUADRUPLE
2-INPUT POSITIVE
NAND GATES



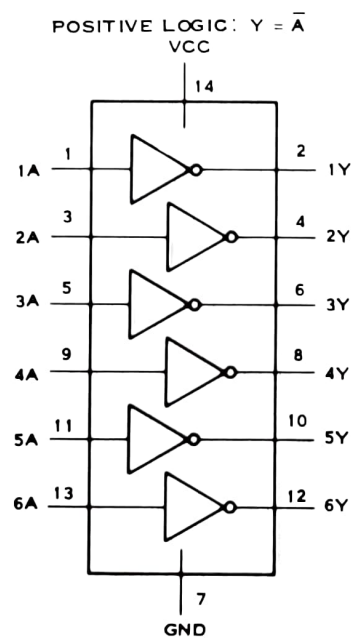
SN7402

QUAD 2-INPUT
POSITIVE NOR
GATES



SN74H04

HEX INVERTERS

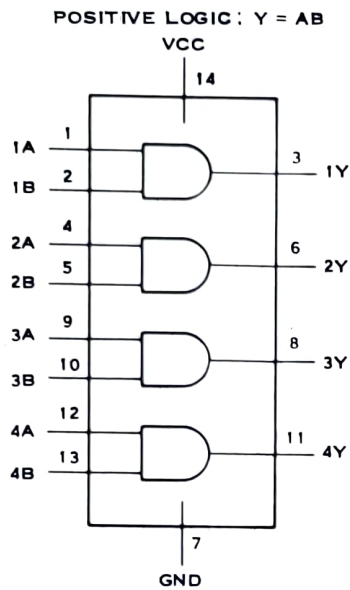


SN7405

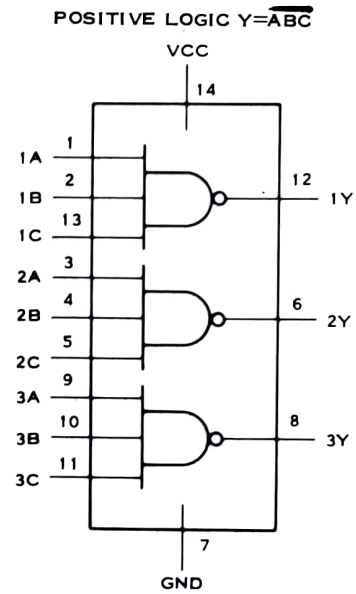
HEX INVERTERS
(WITH OPEN-
COLLECTOR OUTPUT)

(B)128298 (1/8)

Figure 7-1. Integrated Circuit Diagrams (Sheet 1 of 8)

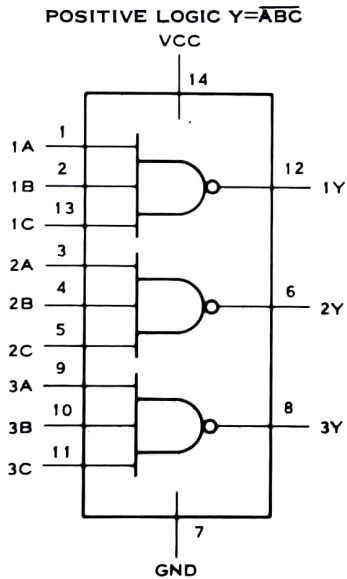


SN7408
QUAD 2-INPUT
POSITIVE AND
GATES



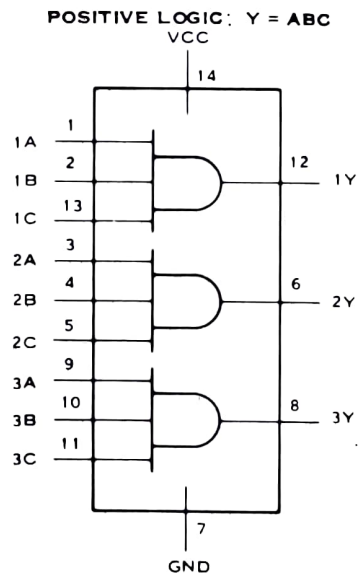
SN7410

TRIPLE 3-INPUT
POSITIVE
NAND GATES



SN74H10

TRIPLE 3-INPUT
POSITIVE
NAND GATES

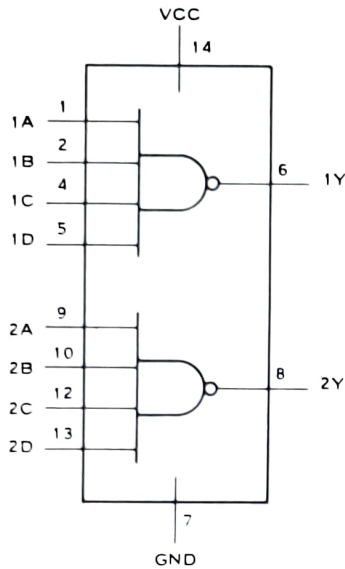


SN74H11

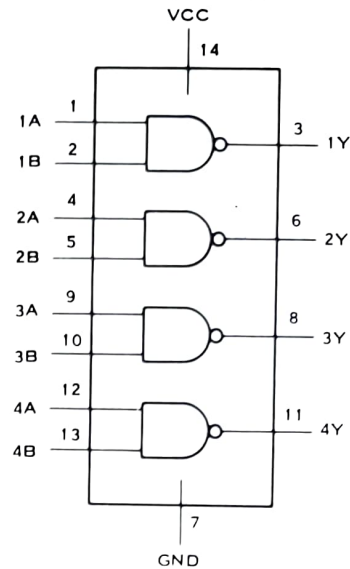
TRIPLE 3-INPUT
POSITIVE AND
GATES

(B)128298 (2/8)

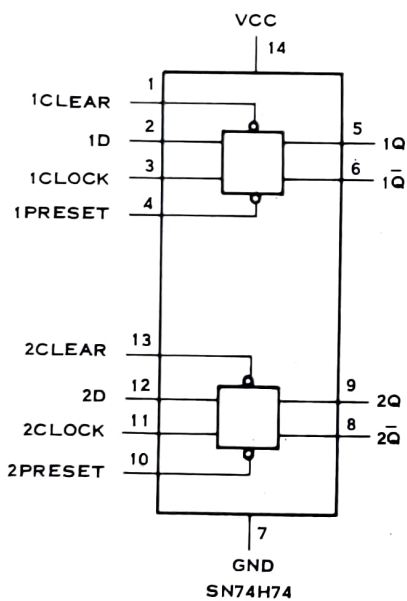
Figure 7-1. Integrated Circuit Diagrams (Sheet 2 of 8)

POSITIVE LOGIC: $Y = \overline{ABCD}$ 

SN7420
DUAL 4- INPUT
POSITIVE NAND
GATES

POSITIVE LOGIC $Y = \overline{AB}$ 

SN7437
QUAD 2- INPUT
POSITIVE NAND
GATES AND BUFFERS

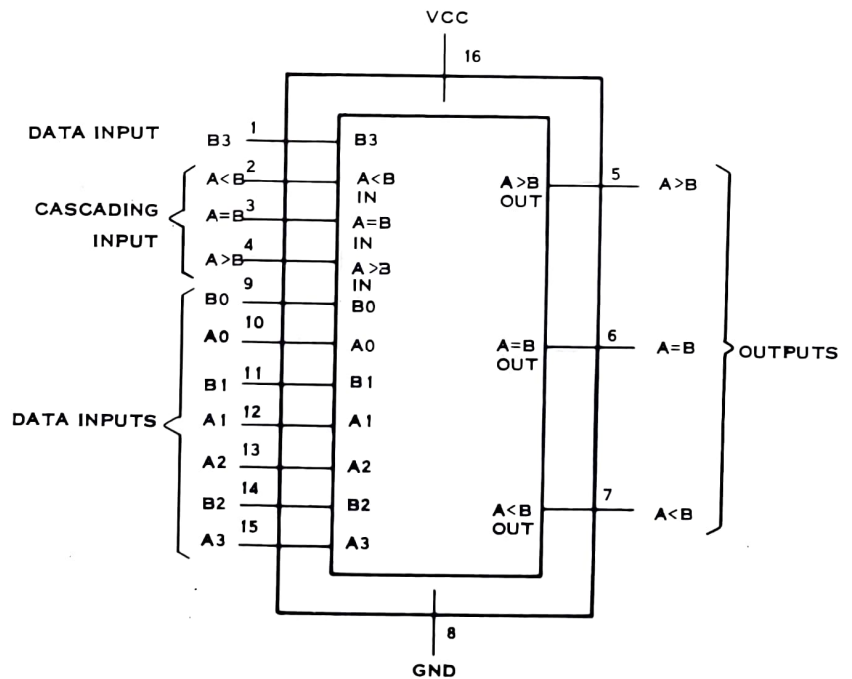


SN74H74

D-TYPE EDGE-TRIGGERED FLIP-FLOPS

TRUTH TABLE		
D	Q	Q
L	L	H
H	H	L

(B)128298 (3/8)



SN7485
4-BIT MAGNITUDE COMPARTORS

Figure 7-1. Integrated Circuit Diagrams (Sheet 3 of 8)

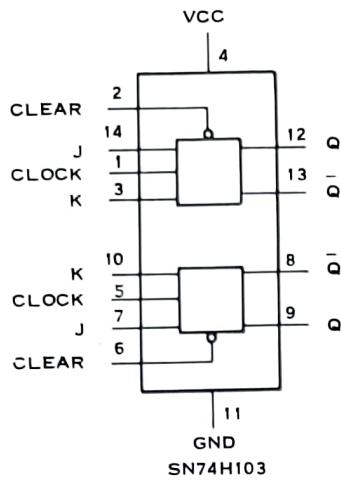


TRUTH TABLE

COMPARING INPUTS				CASCADING INPUTS			OUTPUTS		
A3,B3	A2,B2	A1,B1	A0,B0	A>B	A<B	A=B	A>B	A<B	A=B
A3>B3	X	X	X	X	X	X	H	L	L
A3<B3	X	X	X	X	X	X	L	H	L
A3=B3	A2>B2	X	X	X	X	X	H	L	L
A3=B3	A2<B2	X	X	X	X	X	L	H	L
A3=B3	A2=B2	A1>B1	X	X	X	X	H	L	L
A3=B3	A2=B2	A1<B1	X	X	X	X	L	H	L
A3=B3	A2=B2	A1=B1	A0>B0	X	X	X	H	L	L
A3=B3	A2=B2	A1=B1	A0<B0	X	X	X	L	H	L
A3=B3	A2=B2	A1=B1	A0=B0	H	L	L	H	L	L
A3=B3	A2=B2	A1=B1	A0=B0	L	H	L	L	H	L
A3=B3	A2=B2	A1=B1	A0=B0	L	L	H	L	L	H

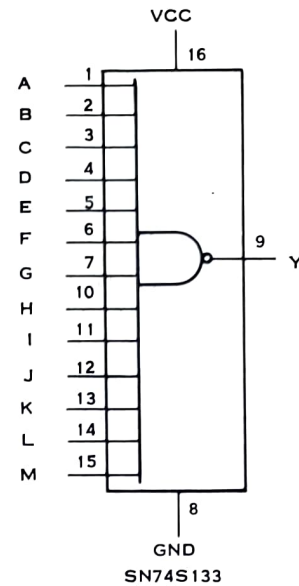
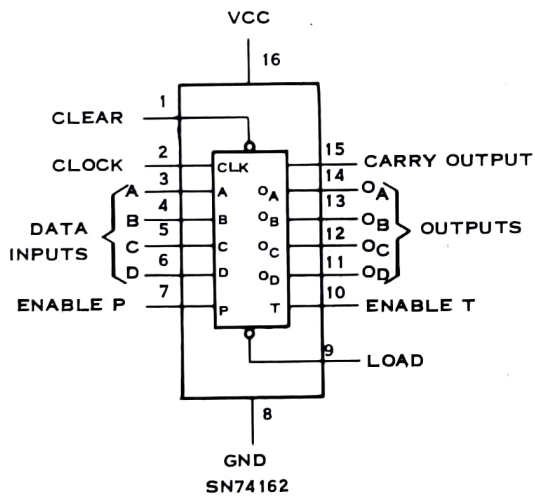
NOTE: H = HIGH LEVEL, L = LOW LEVEL, X = IRRELEVANT

(A)128298 (4/8)

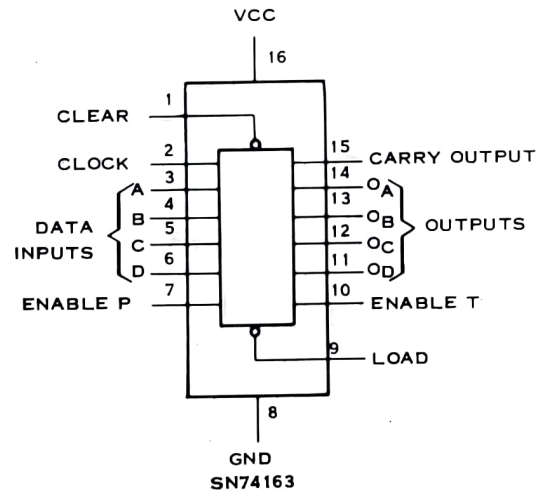


J - K EDGE-TRIGGERED FLIP-FLOP

TRUTH TABLE		
J	K	Q
0	0	Q
0	1	0

POSITIVE-NAND GATES AND INVERTERS WITH TOTEM-POLE OUTPUTS
POSITIVE LOGIC $Y = \overline{ABCDEFGHIJKLM}$ 

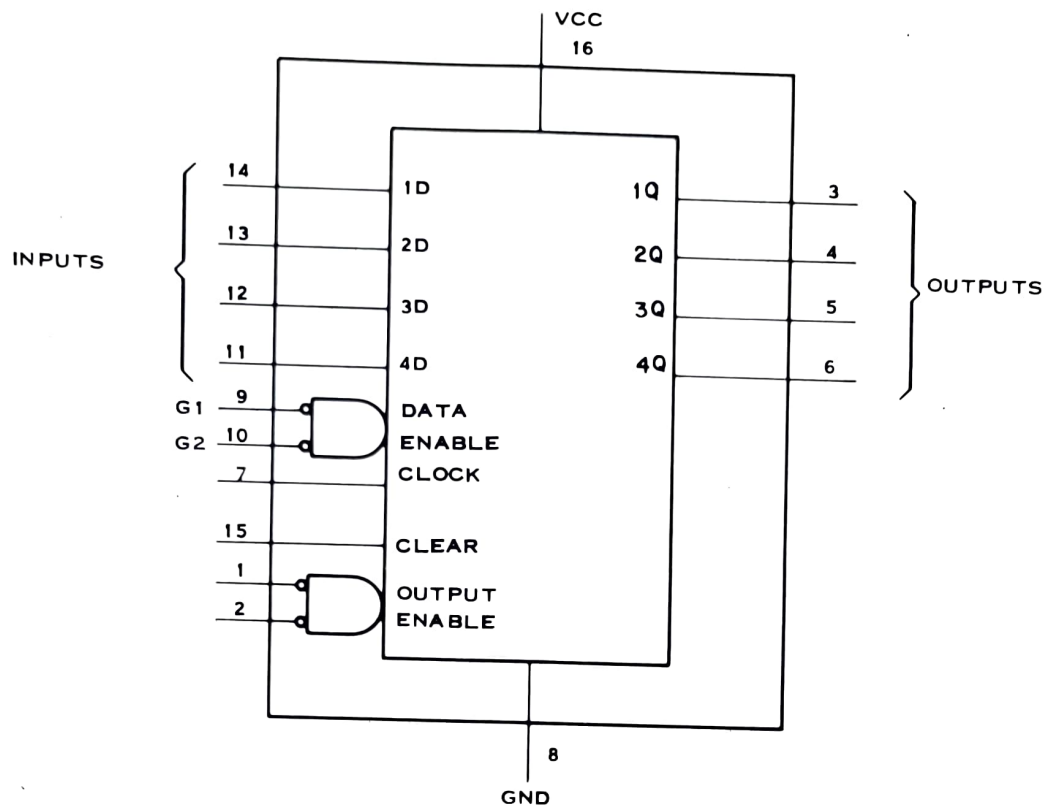
SYNCHRONOUS FOUR-BIT COUNTER



SYNCHRONOUS FOUR-BIT COUNTER

(B)128298 (5/8)

Figure 7-1. Integrated Circuit Diagrams (Sheet 5 of 8)



SN74173
4-BIT D-TYPE REGISTERS
WITH 3-STATE OUTPUTS

FUNCTION TABLE

FUNCTION TABLE					
INPUTS					OUTPUT Q
CLEAR	CLOCK	DATA ENABLE		DATA D	
		G1	G2		
H	X	X	X	X	L
L	L	X	X	X	Q ₀
L	↑	H	X	X	Q ₀
L	↑	X	H	X	Q ₀
L	↑	L	L	I	L
L	↑	L	L	H	H

WHEN EITHER M OR N (OR BOTH) IS (ARE) HIGH THE OUTPUT IS DISABLED TO THE HIGH IMPEDANCE STATE; HOWEVER, SEQUENTIAL OPERATION OF THE FLIP FLOPS IS NOT AFFECTED.

H - HIGH LEVEL (STEADY STATE)

L - LOW LEVEL (STEADY STATE)

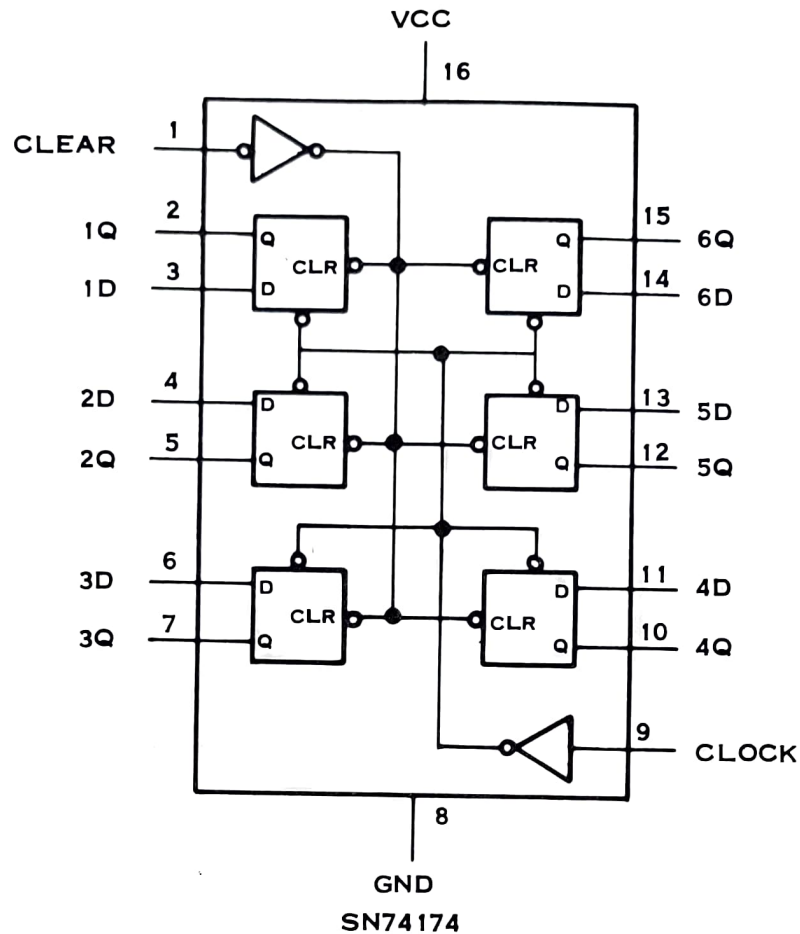
↑ - LOW TO HIGH LEVEL TRANSITION

X - IRRELEVANT (ANY INPUT INCLUDING TRANSITIONS)

Q₀ - THE LEVEL OF Q BEFORE THE INDICATED STEADY STATE INPUT CONDITIONS WERE ESTABLISHED.

(A)128298 (6/8)

Figure 7-1. Integrated Circuit Diagrams (Sheet 6 of 8)

**D-TYPE FLIP-FLOP WITH CLEAR****FUNCTION TABLE
(EACH FLIP-FLOP)**

INPUTS			OUTPUT
CLEAR	CLOCK	D	Q
L	X	X	L
H	↑	H	H
H	↑	L	L
H	L	X	Q ₀

H = HIGH LEVEL (STEADY STATE)

L = LOW LEVEL (STEADY STATE)

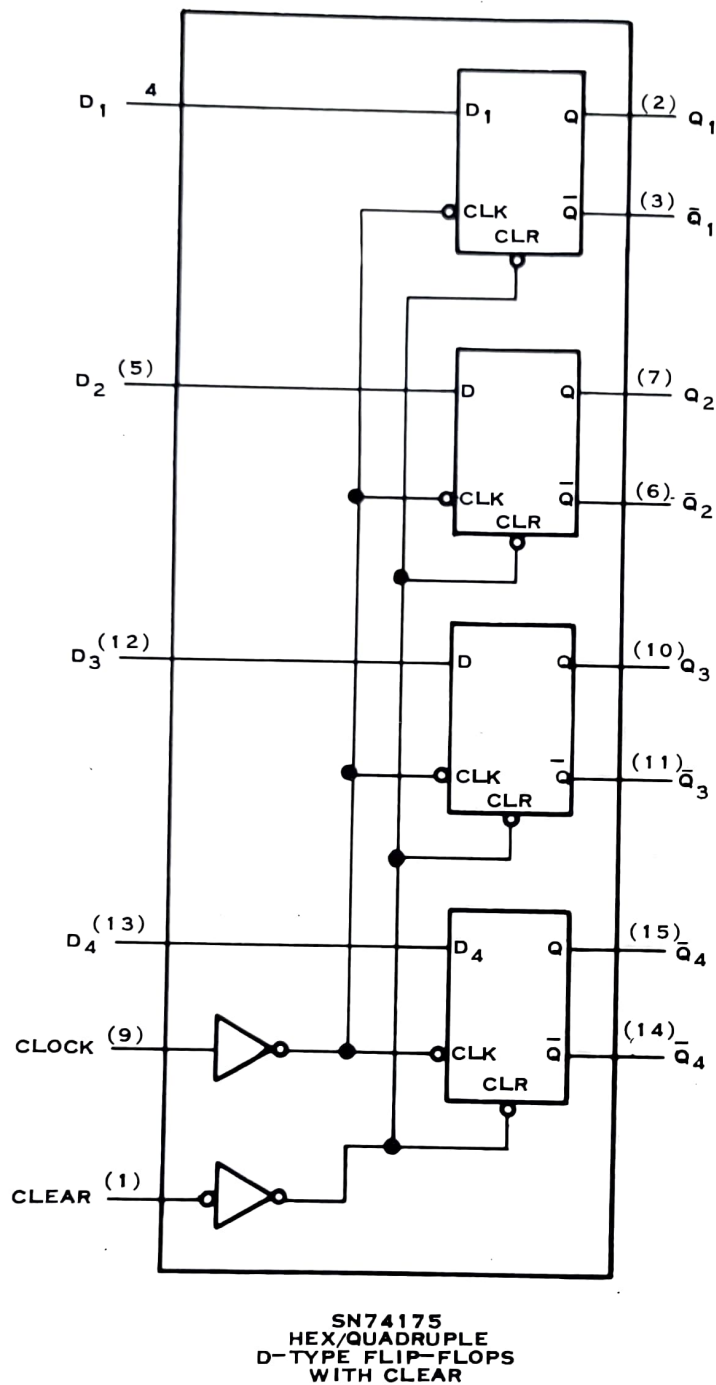
X = IRRELEVANT

↑ = TRANSITION FROM LOW TO HIGH

Q₀ = THE LEVEL OF Q BEFORE THE INDICATED STEADY-STATE INPUT CONDITIONS WERE ESTABLISHED.

(A)128298 (7/8)

Figure 7-1. Integrated Circuit Diagrams (Sheet 7 of 8)



FUNCTION TABLE

INPUTS			OUTPUTS	
CLEAR	CLOCK	D	Q	$\bar{Q}$
L	X	X	L	H
H	↑	H	H	L
H	↑	L	L	H
H	L	X	Q_0	$\bar{Q}_0$

H = HIGH LEVEL (STEADY STATE)

L = LOW LEVEL (STEADY STATE)

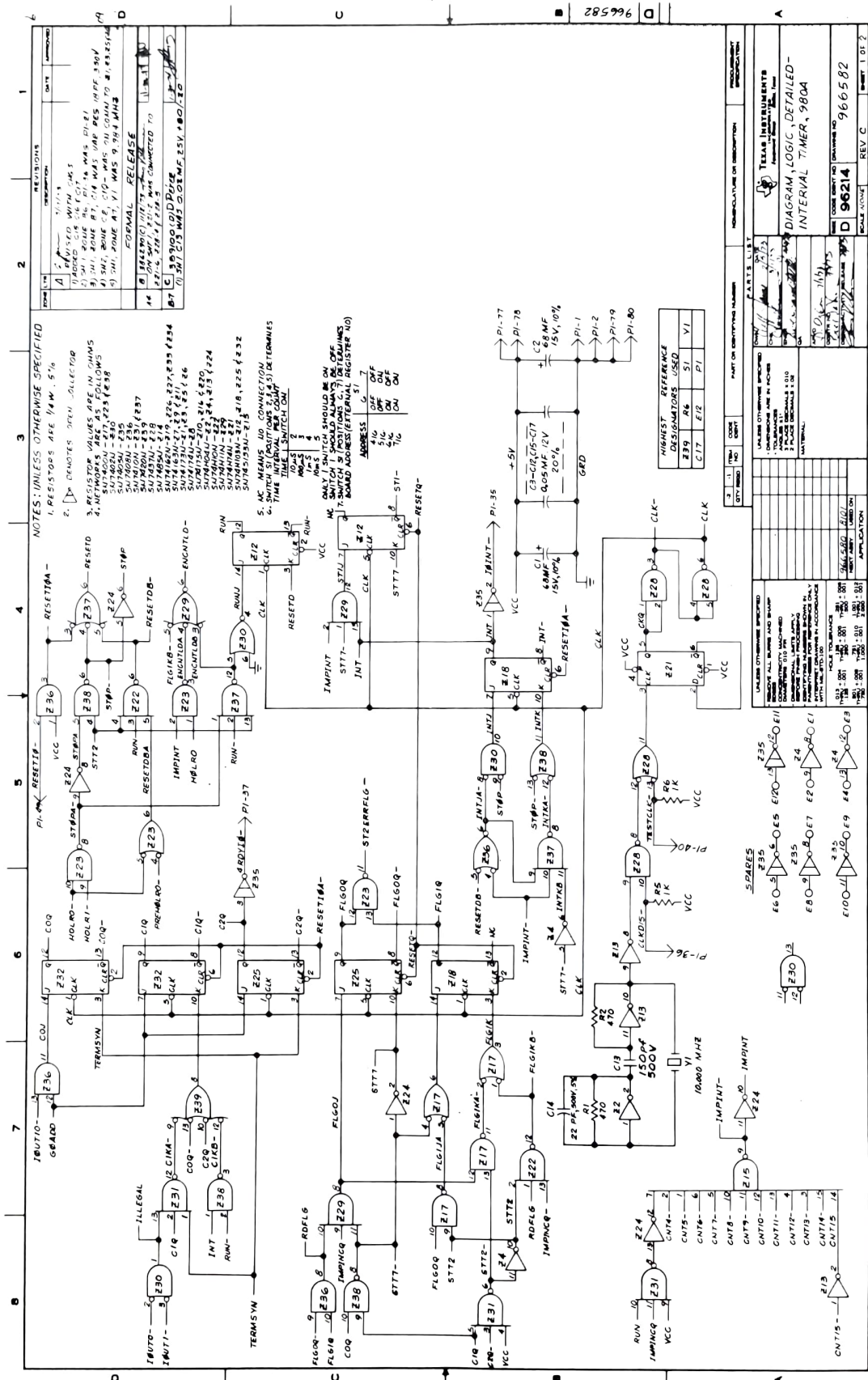
X = IRRELEVANT

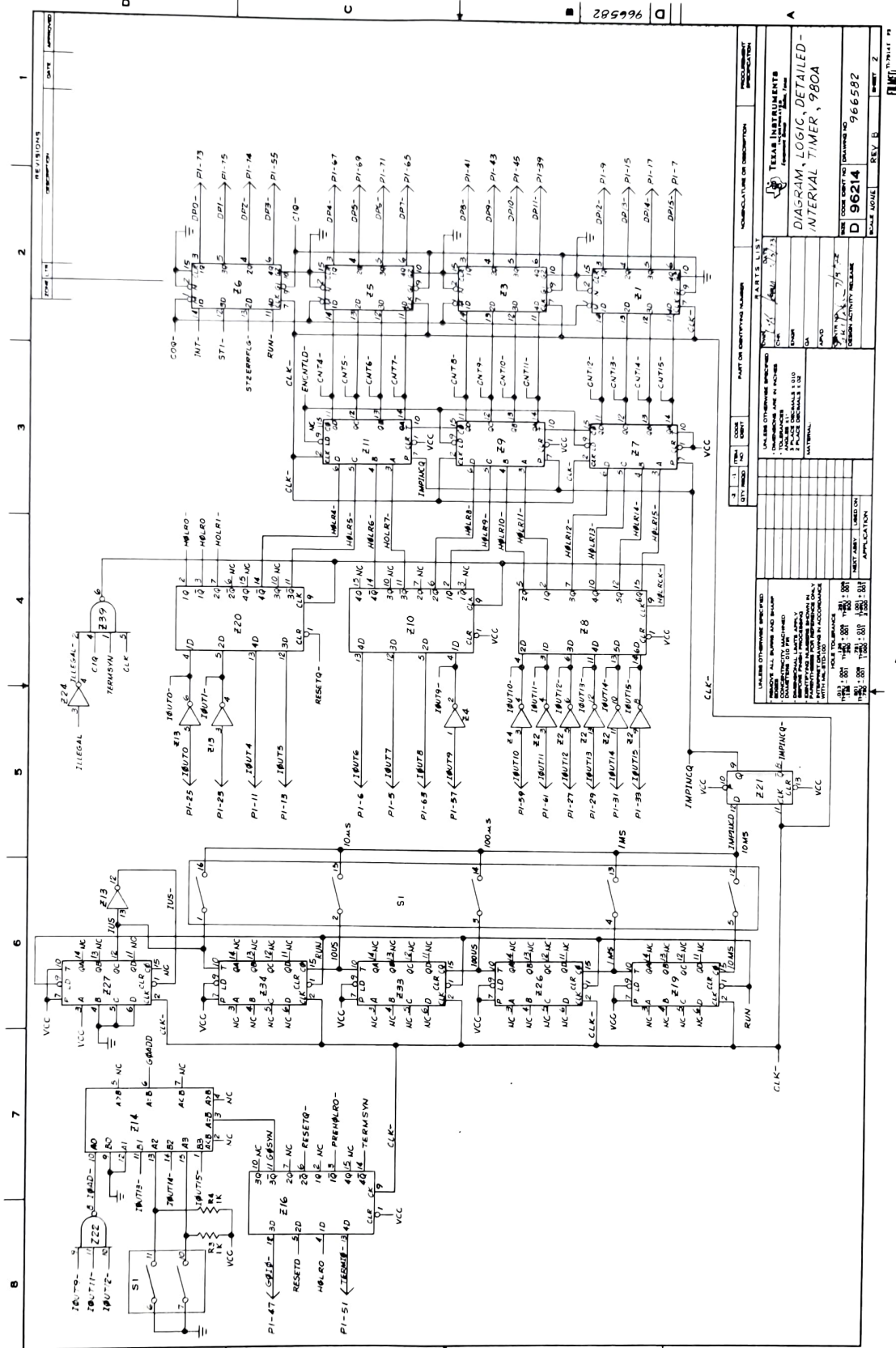
↑ = TRANSITION FROM LOW TO HIGH

 Q_0 = THE LEVEL OF Q BEFORE THE INDICATED STEADY-STATE INPUT CONDITIONS WERE ESTABLISHED.

(A)128298 (8/8)

Figure 7-1. Integrated Circuit Diagrams (Sheet 8 of 8)







APPENDIX A
GLOSSARY OF SIGNAL NAMES



APPENDIX A
GLOSSARY OF SIGNAL NAMES

CLK	CLock
CLKDIS	CLock DISable
CNT4-15	CouNTER register bits 4 through 15
ENCNTLD	ENable CouNTER register Load signal
GOADD	GOod ADDRESS
HOLR0-15	HOLding Register bits 0 through 15
ILLEGAL	ILLEGAL function
IMPINC	IMPending INCrement
IMPINT	IMPending INTerrupt
INT	INTerrupt
IOAD	Input/Output ADDRESS
IOINT	Input/Output INTerrupt signal
IOUT0-15	Input/OUTput bits 0 through 15
PREHOLR0	PREvious HOLding Register bit 0
RDFLG	ReaD FLA G
RDYIO	ReaDY Input/Output
RUN	RUN mode
RUN-	Not RUN mode
STT2	STate Time 2
STT7	STate Time 7
ST2ERRFLG	STatus 2 ERRor FLA G
TESTCLK	TESTCLock





943001-9701

ALPHABETICAL INDEX



ALPHABETICAL INDEX

INTRODUCTION

The following index lists key words and concepts from the subject material of the manual together with the area(s) in the manual that supply major coverage of the listed concept. The numbers along the right side of the listing reference the following manual areas:

- Sections - References to Sections of the manual appear as "Section x" with the symbol x representing any numeric quantity.
- Appendixes - References to Appendixes of the manual appear as "Appendix y" with the symbol y representing any capital letter.
- Paragraphs - References to paragraphs of the manual appear as a series of alphanumeric or numeric characters punctuated with decimal points. Only the first character of the string may be a letter; all subsequent characters are numbers. The first character refers to the section or appendix of the manual in which the paragraph is found.
- Tables - References to tables in the manual are represented by the capital letter T followed immediately by another alphanumeric character (representing the section or appendix of the manual containing the table). The second character is followed by a dash (-) and a number:

Tx-yy

- Figures - References to figures in the manual are represented by the capital letter F followed immediately by another alphanumeric character (representing the section or appendix of the manual containing the figure). The second character is followed by a dash (-) and a number:

Fx-yy

- Other entries in the Index - References to other entries in the index are preceded by the word "See" followed by the referenced entry.



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7

8



9

10



USER'S RESPONSE SHEET

Manual Title: Operation and Maintenance Instructions for

Model 980 Interval Timer (943001-9701)

Manual Date: 1 July 1975 Date of This Letter: _____

User's Name: _____ Telephone: _____

Company: _____ Office/Department: _____

Street Address: _____

City/State/Zip Code: _____

Please list any discrepancy found in this manual by page, paragraph, figure, or table number in the following space. If there are any other suggestions that you wish to make, feel free to include them. Thank you.

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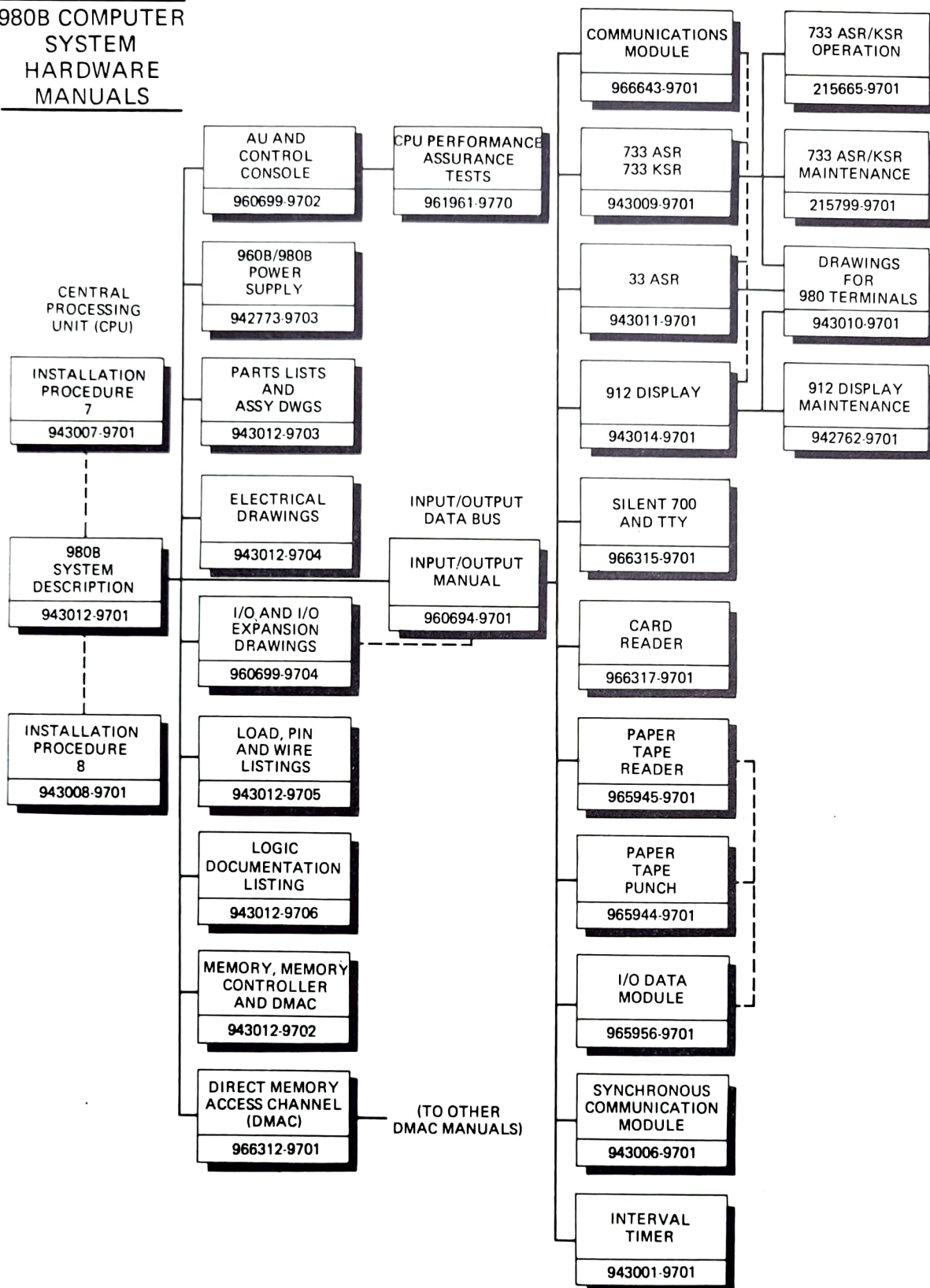


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980B COMPUTER SYSTEM HARDWARE MANUALS



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